

Engineering Architecture & Service Guidelines

Microservice Infrastructure Overview

Our core platform relies on Python 3.11 with Django REST Framework and FastAPI for sub-millisecond asynchronous execution. Asynchronous workloads such as vector indexing, embedding calculations, and background PDF ingestion are orchestrated using Celery workers coupled with Redis as a message broker. All microservices are containerized using multi-stage Docker images and fronted by an Nginx reverse proxy.

Vector Database & Embedding Specifications

For dense semantic retrieval, the platform implements FAISS CPU IndexIDMap utilizing 384-dimensional dense vectors generated via sentence-transformers (all-MiniLM-L6-v2). Lexical keyword retrieval is handled concurrently by BM25Plus, which eliminates the zero-IDF saturation artifact on small corporate corpora. Both dense and sparse ranks are merged using Reciprocal Rank Fusion (RRF) with constant k=60, before passing the top 20 candidates to the Cross-Encoder re-ranker.

Service Latency and Availability SLAs

Service Name	Target TTFT	P95 Latency	Availability SLA
RAG Streaming SSE	45 ms	120 ms	99.95%
Hybrid Vector Retrieval	12 ms	28 ms	99.99%
Cross-Encoder Reranker	35 ms	70 ms	99.90%
Background PDF Ingestion	N/A (Async)	8.5 sec	99.50%